

PROJECT REPORT

Automated Network Request Management in ServiceNow

Final Report

Parameter	Details
Project Name	Automated Network Request Management in ServiceNow
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1. INTRODUCTION

1.1 Project Overview

Automated Network Request Management in ServiceNow is a service-management solution designed to centralize and automate the process of requesting network access, VPN access, firewall rules, and network connectivity. The solution provides a Service Catalog item through the Service Portal, uses dynamic form behavior and client-side validation, and automates approval and fulfillment using Flow Designer.

The implemented solution uses a custom Network Request table, Service Catalog variables, requester information auto-population, manager approval, Network Team fulfillment tasks, request-state tracking, email notifications, reports, and a Platform Analytics dashboard.

1.2 Purpose

- Provide a simple and centralized way for employees to submit network requests.
- Display request-specific information dynamically and validate important technical inputs.
- Reduce manual coordination by automating manager approval and fulfillment routing.
- Provide clear request-state visibility to requesters, approvers, and the Network Team.
- Provide reports and a dashboard for monitoring operational request information.

2. IDEATION PHASE

2.1 Problem Statement

How might we automate the network request management process so that users can easily submit requests, managers can efficiently approve or reject them, and the Network Team can fulfill approved requests?

2.2 Empathy Map Canvas

Empathy Area	Project Findings
Persona	Network Requester / Employee
Goal	Make the network request process simple, fast, and transparent.
Hear	Requests for required network details, manager approval, and Network Team processing.
See	ServiceNow Service Portal, Network Request form, dynamic fields, request status, notifications.
Say	I need network access for my work; I want the form to be simple and I want to know my status.
Do	Select request type, enter details, submit request, check approval/status.
Think & Feel	Wants a quick submission and is concerned about incorrect information and approval delays.
Pains	Manual steps, unclear required information, incorrect input, approval delays, limited visibility.
Gains	Dynamic form, validation, automated approval, faster fulfillment, clear status visibility.

2.3 Brainstorming

The brainstorming activity grouped ideas into request submission, form and validation, approval and fulfillment, notifications, and monitoring/security.

Idea Group	Ideas
Request Submission	Service Catalog item, Service Portal, capture request details, create request record.
Form & Validation	Dynamic fields, requester auto-population, IP validation, port validation.
Approval & Fulfillment	Manager approval, rejection handling, Network Team task, state tracking.
Notifications	Approval, rejection, and completion email notifications.
Monitoring & Security	Reports, dashboard, roles, groups, controlled access.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Customer Journey
Entice	Requester learns that network services can be requested through ServiceNow.
Enter	Requester opens the Service Portal, selects Network Request, completes the form, and submits it.
Engage	Manager reviews the approval; approved requests are routed to the Network Team.
Exit	Requester receives the outcome and can see whether the request is Completed or Rejected.
Extend	Requester can review request history and use the Service Catalog for future requests.

3.2 Solution Requirement

No.	Functional Requirement	Key Sub-Requirements
FR-1	Network Request Submission	Submit through Service Catalog; create RITM; capture request information.
FR-2	Dynamic Form & Validation	Request-type dependent fields; requester auto-population; firewall IP/port validation.
FR-3	Approval & Fulfillment	Manager approval; rejection path; Network Team catalog task; state tracking.
FR-4	Notifications & Monitoring	Email notifications; reports; dashboard for operational monitoring.
NFR	Requirement	Description
NFR-1	Usability	Simple, user-friendly catalog experience with relevant dynamic fields.
NFR-2	Security	Role and group based access for requesters, approvers, fulfillers, and administrators.
NFR-3	Reliability	Consistent request, approval, and fulfillment processing.
NFR-4	Performance	Efficient submission, validation, and workflow processing.
NFR-5	Availability	Service available to authorized users through ServiceNow.
NFR-6	Scalability	Support additional request types, rules, groups, and integrations.

3.3 Data Flow Diagram

The data flow begins with the requester and Service Portal, creates a Requested Item (RITM), and passes catalog information to Flow Designer. The flow creates the Network Request record, requests manager approval, and routes approved work to the Network Team through a Catalog Task.

Source	Process	Destination / Data
Requester	Service Catalog submission	Network Request Catalog Item / RITM
RITM	Flow Designer processing	Catalog variables / Network Request record
Network Request	Manager approval	Approval request and approval state
Approved request	Fulfillment	Catalog Task assigned to Network Team
Task completion	State update	Network Request becomes Completed
Rejected request	Rejection path	Network Request becomes Rejected

3.4 Technology Stack

Technology / Component	Usage
ServiceNow Platform	Main platform for service management and application configuration.
Service Catalog	Catalog item and request submission.
Service Portal	User-facing request submission interface.
Flow Designer	Workflow automation, approval, decision logic, and fulfillment task creation.
Custom Table	u_network_request stores the Network Request information.

Requested Item (sc_req_item)	Stores the catalog-generated RITM.
Catalog Task	Represents fulfillment work assigned to Network Team.
Catalog UI Policies	Controls dynamic visibility and mandatory behavior.
Catalog Client Scripts	Performs firewall IP and port validation.
Platform Analytics	Reports/dashboard monitoring.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Canvas Area	Project Fit
Customer Segment(s)	Employees/requesters, managers/approvers, and Network Team members.
Customer Constraints	Different request types need different information; approval and fulfillment can require coordination.
Available Solutions	Email/manual requests, basic forms, spreadsheets/manual tracking, existing service processes.
Jobs-to-be-Done / Problems	Submit correctly, obtain approval, get network service fulfilled, track status.
Problem Root Cause	Multiple request requirements and people involved without a single automated process.
Behaviour	Users submit requests, managers approve/reject, Network Team fulfills approved requests.
Triggers	Need for access, VPN, firewall rule, connectivity, or a change to existing access.
Emotions	Before: confused/concerned/frustrated. After: informed/confident/in control.
Your Solution	Automated Network Request Management in ServiceNow.
Channels	Online: Service Portal, Catalog, email. Offline: workplace coordination with managers/Network Team.

4.2 Proposed Solution

The proposed solution centralizes network requests in ServiceNow. A requester selects the Network Request catalog item and provides the required information. The form dynamically displays fields based on request type and validates firewall inputs. The submitted catalog request creates an RITM and triggers the automated processing flow. The flow creates the Network Request record, asks for manager approval, and either rejects the request or creates a fulfillment task for the Network Team. After fulfillment, the request is marked Completed.

4.3 Solution Architecture

Component	Architecture Role
Requester	Initiates a network request.
Service Portal	Provides the request submission interface.
Network Request Catalog Item	Captures structured network request information.
RITM	ServiceNow Requested Item record created from the catalog order.
Flow Designer	Coordinates data retrieval, record creation, approval, decision, and fulfillment.
u_network_request	Custom table storing the processed network request.
Manager Approval	Controls whether the request proceeds.
Network Team Catalog Task	Provides the fulfillment work item.
Notifications	Communicates important request events.
Reports & Dashboard	Provides monitoring and operational visibility.

End-to-end flow:

1. Requester submits Network Request from Service Portal.
2. ServiceNow creates the RITM.
3. Flow Designer retrieves catalog variables and creates the Network Request record.
4. Manager reviews and approves or rejects the request.
5. Approved requests move to In Progress and create a Network Team Catalog Task.
6. Network Team completes the task and the request moves to Completed.
7. Rejected requests move to Rejected.

8. Notifications and dashboard/reporting provide visibility.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Epic / Functional Area	Story Points
Sprint-1	Network Request Submission	6
Sprint-1	Dynamic Form & Validation	8
Sprint-2	Approval & Fulfillment	15
Sprint-2	Request Tracking	2
Sprint-3	Notifications & Monitoring	7
Sprint-4	Security & Testing/Deployment	5

The planning logic uses the reference Fibonacci-style story point scale of 1, 2, 3, and 5, where higher values represent greater estimated effort. Sprint allocation can be updated with actual Jira completion data.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing for this ServiceNow solution focuses on practical response and processing behavior rather than high-volume benchmark testing. The following checks are recommended for the implemented PDI environment.

Test ID	Test	Expected Result
PT-01	Open Network Request catalog item	Form loads successfully.
PT-02	Change Request Type	Relevant dynamic fields appear without unnecessary fields.
PT-03	Submit valid request	RITM is created and processing starts.
PT-04	Submit invalid firewall IP	Client-side validation prevents submission.
PT-05	Submit invalid firewall port	Client-side validation prevents submission.
PT-06	Manager approval	Approval action completes and flow follows approved path.
PT-07	Manager rejection	Flow follows rejection path and request becomes Rejected.
PT-08	Approved fulfillment	Catalog Task is created and assigned to Network Team.
PT-09	Task completion	Request state progresses to Completed.
PT-10	Dashboard load	Configured visualizations load request monitoring data.

7. RESULTS

7.1 Output Screenshots

Insert the actual screenshots from the implemented ServiceNow instance in the following order. The final report template requires an Output Screenshots section; the screenshots should represent the tested solution.

Figure	Screenshot to Insert
Figure 1	Network Request Catalog Item in Service Portal
Figure 2	Network Request catalog form with dynamic fields
Figure 3	RITM created after submitting the request
Figure 4	Network Request record in the custom table
Figure 5	Manager Approval record / My Approvals
Figure 6	Approved request and Network Team Catalog Task
Figure 7	Rejected request result
Figure 8	Email notification / email log
Figure 9	Reports – Network Requests by State/Type/Priority/Device/Environment
Figure 10	Network Request Management Dashboard

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
Centralized network request submission.	Requires ServiceNow platform access.
Dynamic fields reduce unnecessary form input.	Initial configuration requires administrator effort.
Client-side validation reduces common input errors.	Current solution uses configured rules and may need expansion for more network scenarios.
Automated manager approval and fulfillment.	Approval depends on the availability of the assigned manager.
Clear request state and operational visibility.	PDI testing does not represent production-scale performance.
Reports and dashboard support monitoring.	Additional integrations are needed for direct network-device execution.

9. CONCLUSION

The Automated Network Request Management solution demonstrates how ServiceNow can be used to convert a multi-step network request process into a structured and automated service workflow. The implementation provides a Service Catalog entry point, dynamic request behavior, validation, manager approval, Network Team fulfillment, state tracking, notifications, and monitoring. The resulting design improves process consistency and visibility while providing a foundation that can be extended with additional request types and integrations.

10. FUTURE SCOPE

- Integrate with network security and network-device systems for automated rule/access provisioning.
- Add more advanced approval matrices based on access level, environment, priority, and security impact.
- Add stronger technical validation for IP addresses, ports, protocols, and network ranges.
- Add automated status updates from downstream network systems.
- Add richer operational analytics and trend reporting.
- Introduce additional notifications and escalation mechanisms.
- Expand the solution for additional network request categories and departments.

11. APPENDIX

Source Code (if any)

The implementation primarily uses ServiceNow configuration, Flow Designer, Catalog UI Policies, and Catalog Client Scripts. The following client-side scripts were used for firewall input validation.

Dataset Link

Not applicable. This project does not use an external dataset.

GitHub & Project Demo Link

Not provided in the supplied project materials. Add the GitHub repository and project demo URL here if available.

Report Basis

This report follows the section order in the supplied Final Report Template: Introduction, Ideation Phase, Requirement Analysis, Project Design, Project Planning & Scheduling, Functional and Performance Testing, Results, Advantages & Disadvantages, Conclusion, Future Scope, and Appendix. The template lists these sections on page 1.